The chart posterior limit

Claude, with Daniel Murfet

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Abstract

Astra #13 unit u159, closing the posterior-ratio programme at chart level. For a common random phase converging in distribution and deterministic amplitudes y_φ, y_1 with $y_{1,00} > 0$, the chart posterior ratio $Z_{N_n}[\varphi]/Z_{N_n}[1]$ converges in distribution to the deterministic corner-value ratio $y_{\varphi,00}/y_{1,00}$ (`tendstoInDistribution_chart_posterior`). The normalised pair converges jointly (unit 158), the limiting denominator is positive (unit 154), so the quotient of normalised chart integrals converges (unit 157); the common factor $\int_0^\infty s^{p-1}e^{-\beta s^2+\beta s x_{00}}ds$ cancels in the limit (`coeffA_withY_div`) and the normalisation s_n cancels in the quotient for $N_n > 1$. Lean's totalised division is used throughout. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem coeffA_withY_div ... (hy0 : 0 < y  (0,0)) (a : CoeffPair) :
    coeffA    h  h  k  k  p (withY   h  y  hy  a)
      / coeffA    h  h  k  k  p (withY   h  y  hy  a) = y  (0,0) / y  (0,0)
theorem tendstoInDistribution_normA_div ... (hZm : Measurable Z) ... :
    TendstoInDistribution (fun n   => normA … p (Nseq n) (withY … y  … (X n  ))
        / normA … p (Nseq n) (withY … y  … (X n  ))) l
      (fun _ => y  (0,0) / y  (0,0)) (fun _ =>  )  '
theorem normA_div_eq_chartZ_div ... (N) (hN : 1 < N) (a a') :
    normA … p N a / normA … p N a' = chartZ … N a / chartZ … N a'
theorem tendstoInDistribution_chart_posterior (  b p   r T) (h  h  k  k )
    (h  hb hb  hbr hr  hk  hk ) (hp  hp  : equal starts p) (hpT : p < 2 * T)
    (y  y ) (hy  hy ) (hy0 : 0 < y  (0,0))
    (X :   → Ω → CoeffPair) (hXm) (Z : Ω' → CoeffPair) (hZm)
    (hX : TendstoInDistribution X l Z …) (Nseq) (hN : Tendsto Nseq l atTop) :
    TendstoInDistribution (fun n   => chartZ … (Nseq n) (withY … y  … (X n  ))
        / chartZ … (Nseq n) (withY … y  … (X n  ))) l
      (fun _ => y  (0,0) / y  (0,0)) (fun _ =>  )  '
```

2 Mathematics

Both leading coefficients are $\frac{y_{00}}{k_1 k_2}\int_0^\infty s^{p-1}e^{-\beta s^2+\beta s x_{00}}ds$ with the same phase x, so their ratio is $y_{\varphi,00}/y_{1,00}$ (the integral is positive). The quotient theorem of unit 157 applies to the jointly convergent normalised pair with a.s. positive limiting denominator; for $N_n > 1$ the normalisation $N_n^{-p}\log N_n$ is nonzero and cancels in the quotient (the leading exponent has no lower terms), so the quotient of chart integrals and the quotient of normalised chart integrals eventually coincide, which is a null perturbation in probability.

3 Paper-facing meaning

cor:empirical_expectation states that the leading term of the posterior expectation $\mathbb{E}_{w|\mathcal{D}_n}[\varphi] = Z_n[\varphi]/Z_n$ is $A_{i_0,j_0}(\varphi,\xi_n)/B_{k_0,p_0}(\xi_n)$ and converges in distribution. At chart level with equal starting exponents this is now a theorem, and Astra #13's observation makes it sharper than the paper's phrasing: the random fluctuation factor cancels, the limit is the deterministic corner-value ratio $y_{\varphi,00}/y_{1,00}$ (equal to $\varphi(0,0)$ when $\eta_\varphi = \varphi\eta_1$). The empirical fluctuations affect the leading posterior expectation only through the common factor, hence not at all in the limit; they enter at the subleading orders. Caveats (staging): one chart, equal starts, $d = 2$; convergence of the phase data is a hypothesis; the identification $Z_n[\varphi] = \texttt{chartZ}(\texttt{withY}\, y_\varphi\, X_n)$ is the chart-level model of the paper's $Z_n[\varphi]$.

4 Lean notes

- `field_simp` may rewrite inside lambdas under `logMoment` and desynchronise two syntactically equal occurrences (βsx vs βxs); `set M := logMoment …` before `field_simp` folds both into one atom.
- `tendstoInMeasure_of_eventually_abs_le` with a constant zero rate needs the filter passed explicitly (`(l := l)`), since `tendsto_const_nhds` does not determine it.
- `div_div_div_cancel_right hs` for $(a/s)/(b/s) = a/b$ in a field.